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Age	income	student	Credit-rating	buys-computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	excellent	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$\text{Info}(D) = - \sum_{i=1}^m P_i \log_2(P_i)$$

$$\text{Info}_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

$$\text{Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$$

For attribute age

Age	class		I(P,N)
	P N		
<=30	2	3	0.971
31...40	4	0	0
>40	3	2	0.971
	9	5	

$$\begin{aligned} & (2,3) \\ & = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \left(\frac{3}{5}\right) \\ & = 0.971 \end{aligned}$$

$$\begin{aligned} & (4,0) \\ & = -\frac{4}{4} \log_2 \frac{4}{4} - 0 \end{aligned}$$

$$I(4,2) = -\frac{4}{6} \log\left(\frac{4}{6}\right) - \frac{2}{6} \log_2\left(\frac{2}{6}\right)$$

$$= -\frac{4}{6} (-0.585) - \frac{2}{6} (-1.585)$$

$$= 0.39 + 1.0566$$

$$= \cancel{1.4466}$$

$$= 0.918$$

$$I(3,1) = -\frac{3}{4} \log\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right)$$

$$= \cancel{0.84} 0.811$$

$$I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right)$$

$$I(9,5) = 0.910$$

Gain (information) $\neq$ Gain.

$$Info_A(D) = \sum_{j=1}^V \frac{|D_j|}{|D|} \times Info(D_j)$$

$$= \frac{4}{14} \times 1 + \frac{6}{14} \times 0.9174 + \frac{4}{14} \times 0.811$$

$$= 0.910$$

$$\text{Gain} = 0.940 - 0.910$$

$$= 0.030$$

Information Gain:

age	0.246
Income	0.029
Student	0.159
Credit	0.048

ii) Gain ratio

$$\text{Gain Ratio (A)} = \frac{\text{Gain(A)}}{\text{Split Info (A)}}$$

$$\text{Split Info (A)} = - \sum_{j=1}^V \frac{|D_j|}{|D|} \times \log_2 \left(\frac{|D_j|}{|D|} \right)$$

For the attribute age

$$\text{Gain Ratio (A)} = \frac{\text{Gain (age)}}{\text{Split Info (age)}}$$

$$wt = -\frac{5}{14} \log_2\left(\frac{5}{14}\right) - \frac{4}{14} \log_2\left(\frac{4}{14}\right) + \frac{5}{14} \log_2\left(\frac{1}{14}\right)$$

$$\begin{aligned} S_{Plot} &= -\frac{5}{14}(-1.4854) - \frac{4}{14}(-1.8074) + \frac{5}{14}(-1.4854) \\ &= 1.5771 \end{aligned}$$

$$\text{Gain Ratio (Age)} = \frac{\text{Gain (Age)}}{S_{Plot}(\text{Age})}$$

$$= \frac{0.246}{1.5771}$$

$$= 0.156$$

Income:

$$= -\frac{4}{14} \log\left(\frac{4}{14}\right) - \frac{6}{14} \log\left(\frac{6}{14}\right)$$

$$= -\frac{4}{14} \times -1.8074 - \frac{6}{14}(-1.2224)$$

$$= 0.5318 + 0.5238$$

$$= 1.0556$$

$$\text{Gain Ratio} = \frac{0.029}{1.0556}$$

Gain ratio (Income) = $0.186 \approx 0.19$

If an attribute has maximum gain ratio that attribute is taken to be the root node

$$\text{Gain}_A(D) = \frac{|D_1|}{|D|} \text{gini}(D_1) + \frac{|D_2|}{|D|} \text{gini}(D_2)$$

$$\text{gini}(D) = 1 - \sum_{j=1}^n p_j^2$$

$$\begin{aligned} \text{gini}(D) &= 1 - \sum p_j^2 \\ &= 1 - \left[\left(\frac{9}{14} \right)^2 + \left(\frac{5}{14} \right)^2 \right] \end{aligned}$$

$$= 1 - 0.540$$

$$= 0.46$$

D ₁	D ₂
LM-10	H-4
MH	L
LH	M

$$\text{gini}_A(D) = \frac{|D_1|}{|D|} \text{gini}(D_1) + \frac{|D_2|}{|D|} \text{gini}(D_2)$$

$$\begin{aligned} \text{gini}_A(D) &= \left[\frac{10}{14} \left[1 - \left(\frac{7}{10} \right)^2 - \left(\frac{3}{10} \right)^2 \right] \right] + \left[\frac{4}{14} \left[1 - \left(\frac{2}{4} \right)^2 \right] \right] \\ &= 0.443 \end{aligned}$$

2) Generation

Actual class \ Predicted class	Cancer = yes	Cancer = no	Total
Cancer = yes	90	210	300
Cancer = no	140	9560	9700
Total	230	9770	10000

A \ P	C	TC	
C	TP	FN	P
TC	FP	TN	N
	P'	N'	All

i) Sensitivity :

$$\frac{TP}{TP + FN} = \frac{90}{90 + 210} = \frac{90}{300} = 0.3 = 30\%$$

ii) Specificity :

$$\frac{TN}{FP + TN} = \frac{9560}{140 + 9560} = \frac{9560}{9700} = 0.98 = 98.56\%$$

iii) Accuracy:

$$\frac{TP + TN}{TP + FN + FP + TN} = \frac{90 + 9560}{10000} = \frac{9650}{10000} = 0.956 = 95.6\%$$

iv) Precision:

$$\frac{TP}{TP + FP} = \frac{90}{90 + 140} = \frac{90}{230} = 0.391 = 39.1\%$$

$$v) \text{ Recall: } \frac{TP}{TP + FN} = \frac{90 + 210}{90 + 210} = 0.3 = 30\%$$

vi) F-score:

$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.3 \times 0.391}{0.391 + 0.3} = \frac{0.2346}{0.691}$$

$$= 0.3395$$

$$= 33.95\%$$

vii) Error rate:

$$1 - \text{Accuracy} = 1 - 0.956$$

$$1 - 0.956 = 0.044$$

$$= 4.4\%$$

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times Info(D_j)$$

$$Info_{age}(D) = -\frac{5}{14} \times 0.971 + \frac{4}{14} \times 0.915 \times 0.971$$

$$= 0.694$$

$$Gain(Age) = 0.940 - 0.694$$

$$= 0.246$$

Gain(Income):

Income P N

high 2 2

medium 4 2

low 3 1

$$I(2,2) = -\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right)$$

$$= -\frac{2}{4} (-1) - \frac{2}{4} (-1)$$

$$= \frac{2}{4} + \frac{2}{4}$$

$$= 1$$

(3,2)

$$I_{HRO}(0) = \sum_{i=1}^m p_i \log_2(p_i)$$

$$= -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \left(\frac{2}{5}\right)$$

$$= -0.737$$

$$= -\frac{3}{5} \log(-0.737) - \frac{2}{5} (-1.322)$$

$$= 0.971$$

$$I(9,5) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \left(\frac{5}{14}\right)$$

$$= -\frac{9}{14} (-0.6374) - \frac{5}{14} (-1.4854)$$

$$= 0.4097 + 0.5305$$

$$= 0.9402$$